SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Akhila Pochimireddy	**Reg. No.:**	192425225
Date:	08-10-2026	**Duration:**	60 minutes

Code of Conduct

*I **P.Akhila (192425225)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: P.Akhila

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

Problem Explanation

Farmers may find it difficult to reach buyers directly and obtain transparent information about market demand, prices and delivery arrangements. Buyers may also face difficulty finding products, comparing prices and tracking orders.

The Smart Agricultural Marketplace Platform connects farmers, wholesalers, retailers and consumers in one system. Farmers can list products with quantity, quality, price and availability. Buyers can search products, place orders, make secure payments and track deliveries. The system also maintains transactions, ratings, feedback and market analytics.

Objectives

Register farmers and buyers.

2. Manage agricultural product listings.

3. Display product availability and prices.

4. Allow buyers to search and order products.

5. Support secure payments and delivery tracking.

6. Maintain transaction history, ratings and feedback.

7. Provide market and sales analytics.

8. Improve communication and supply-demand coordination.

Scope

- Farmer and buyer registration and profiles
- Product listing, search, filtering, price and availability
- Order, payment and delivery management
- Transaction history, ratings and feedback
- Market analytics and admin management

2. Identify Key Stakeholders and Challenges

Stakeholder	Role	Challenge
Farmers	List products, set price/quantity, manage orders	Finding buyers and price uncertainty
Wholesalers	Purchase products in bulk	Reliable suppliers and price comparison
Retailers	Purchase products for resale	Availability, quality and delivery
Consumers	Purchase agricultural products	Freshness, price and delivery
Delivery Partners	Collect and deliver products	Route and delivery tracking
Payment Gateway	Process online payments	Security and transaction failures
Administrator	Manage users, products and orders	Security, fraud and maintenance
Platform Management	Monitor marketplace performance	Managing transaction/market data

Major Challenges

- Lack of direct communication and transparent price information.
- Dependence on intermediaries and difficulty identifying genuine users.
- Rapid changes in product availability and seasonal demand.
- Perishable products and delivery delays.
- Payment failures or fraudulent transactions.
- Limited technical knowledge among some farmers.

3. Analyze the Current Approach

Traditional process: Farmer → Intermediary/Wholesaler → Retailer → Consumer
Farmers commonly use local markets, wholesalers, retailers or intermediaries to identify buyers and negotiate prices. Demand and price information may not be equally available to all participants.

Strengths of Existing System

- Established local markets and relationships.
- Intermediaries may support transportation.
- Bulk purchasing and physical product inspection are possible.

Limitations

- Multiple intermediaries increase transaction complexity.
- Prices and product availability may not be transparent.
- Manual records can cause errors and order tracking is limited.
- Communication and market-demand information may be slow.
- **Main gap**: no single digital platform for product information, ordering, payment, delivery tracking and analytics.

4. Propose a Suitable Solution Using Software Engineering Principles

Proposed System

The proposed system is a **Smart Agricultural Marketplace Platform** that provides a centralized digital marketplace for farmers, wholesalers, retailers, and consumers.

Major Functional Modules

1. User Registration and Authentication: Registration, login, profiles, authentication and role-based access.
2. Agricultural Product Management: Add product name, quantity, price, quality/details and images; update availability.
3. Marketplace and Product Search: View, search, filter and compare products; check quantity, availability and farmer information.
4. Order Management: Cart, order placement, delivery address, order status and cancellation; farmers accept/reject orders.
5. Payment Management: Secure payment, confirmation, transaction records and payment status.
6. Delivery Tracking: Assign delivery, track progress, update status and confirm delivery.
7. Ratings and Feedback: Buyers rate products/farmers and provide feedback.
8. Market Analytics: Demand, sales, popular products, price trends, order statistics and supply-demand information.

Functional Requirements

1. User registration and authentication.
2. Product listing and updating by farmers.
3. Product search and filtering by buyers.
4. Order placement and management.
5. Secure payment processing.
6. Order and delivery tracking.
7. Transaction history and ratings/feedback.
8. Admin management of users and products.
9. Market and sales analytics.

Non-Functional Requirements

- Security – protect user, payment and transaction information.
- Usability – simple interface for users with limited technical knowledge.
- Scalability – support increasing users, products and transactions.
- Reliability – accurate processing of orders, payments and records.
- Performance – quick product search and order processing.
- Maintainability – independent modules and documented code.
- Availability – system should be accessible when required.

Software Engineering Principles

Modularity	Separate modules for users, products, orders, payments, delivery and analytics
Scalability	Database and architecture designed to support more users
Maintainability	Modular and well-documented code
Reliability	Validation and transaction confirmation

Usability	Simple interface with clear navigation
Security	Authentication, authorization and secure payment processing

5. Justify the Chosen Methodology/Framework

Recommended Methodology: Agile Scrum

The requirements of an agricultural marketplace can change based on feedback from farmers, buyers, delivery partners, and administrators. Therefore, developing the complete system at once may not be suitable.

Example Sprint Plan

- Sprint 1 – Registration and login
- Sprint 2 – Product listing and search
- Sprint 3 – Cart and order management
- Sprint 4 – Payment and transactions
- Sprint 5 – Delivery tracking
- Sprint 6 – Ratings, feedback and analytics
- Sprint 7 – Testing, security and deployment

Scrum Roles

- **Product Owner:** Defines marketplace requirements and priorities.
- **Scrum Master:** Manages the Scrum process and removes obstacles.
- **Development Team:** Designs, develops and tests the platform.
- **Stakeholders:** Farmers, buyers, delivery partners and administrators provide feedback.
- Agile supports changing requirements, continuous feedback, incremental development, early problem identification and continuous testing.

6. Discuss Expected Outcomes and Limitations

Expected Outcomes

1. Improved market access for farmers.
2. Better product visibility and price transparency.
3. Simplified transactions and reduced communication gaps.
4. Better supply-demand coordination.
5. Faster order processing and easier delivery tracking.
6. Digital transaction records and improved market insights.

Potential Risks and Limitations

- Limited digital literacy and rural internet connectivity.
- Product quality may differ from the listed description.
- Payment failures or fraudulent transactions.
- Delivery delays affecting perishable products.
- Difficulty maintaining real-time availability.
- Initial implementation and maintenance costs.

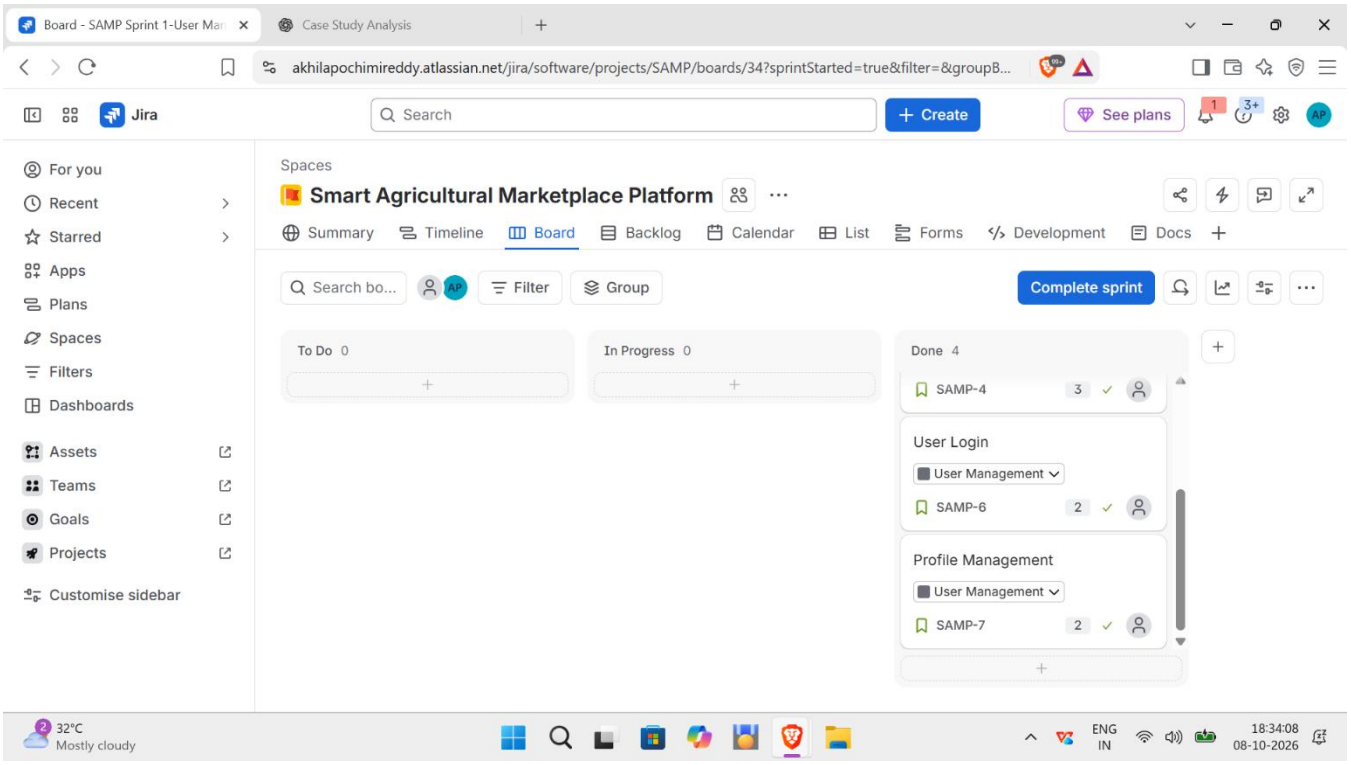
Future Enhancements

- AI-based demand and crop price prediction.
- Smart product recommendations.
- Multilingual and voice-based product listing.
- Location-based farmer-buyer matching.
- IoT-based storage/temperature monitoring.

Blockchain transaction traceability and government-service integration.

Conclusion

The Smart Agricultural Marketplace Platform provides a centralized digital solution connecting farmers, wholesalers, retailers and consumers. It addresses limited market access, price transparency, communication gaps, and order/delivery difficulties. Agile Scrum enables incremental development and stakeholder feedback, while modular, scalable, secure, reliable and user-friendly design supports efficient agricultural transactions.



Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System	Thoroughly	Good analysis	Limited analysis	Proposed solution is

Analysis (20%)	analyzes the existing system, strengths, weaknesses, and identifies all gaps.	with most strengths and weaknesses identified.	with partial identification of issues.	unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25